

FISHER BRANCH, MB, Canada

WMO: 714420

Lat: 51.0834N

Lon: 97.5544W

Elev: 253

StdP: 98.32

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-30.6	-40.6	0.1	-31.3	-38.4	0.1	-30.3	9.6	-13.5	8.9	-13.7	2.0	290	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	29.3	20.3	27.6	19.4	26.0	18.3	22.0	27.0	20.7	25.6	19.6	24.4	4.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.4	25.0	18.9	14.1	23.6	17.7	13.1	22.3	65.7	26.9	61.1	25.6	57.1	24.5	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.0	7.1	-37.7	32.5	3.0	2.1	-39.9	34.0	-41.6	35.2	-43.3	36.4	-45.5	38.0		
(5)			-37.4	24.2	2.5	1.5	-39.2	25.3	-40.7	26.2	-42.1	27.0	-43.9	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-17.6	-15.1	-7.0	2.6	9.8	15.7	18.6	17.5	12.3	4.6	-5.0	-13.8		
	DBStd	14.05	8.19	7.67	7.89	6.38	5.00	3.94	3.25	3.44	4.48	4.80	6.75	7.90		
	HDD10.0	3776	854	702	527	229	67	4	0	0	26	179	449	738		
	HDD18.3	6070	1113	935	785	471	268	96	37	58	186	426	699	996		
	CDD10.0	850	0	0	1	9	59	176	266	232	96	12	0	0		
	CDD18.3	102	0	0	0	0	3	17	45	32	6	0	0	0		
	CDH23.3	1119	0	0	0	2	61	193	439	349	71	3	0	0		
	CDH26.7	242	0	0	0	0	13	36	96	84	13	1	0	0		
(14) Wind	WSAvg	3.3	3.2	3.2	3.4	3.6	3.7	3.2	2.8	3.0	3.3	3.6	3.4	3.2		
(15) Precipitation	PrecAvg	553	30	18	34	33	63	85	66	64	57	42	32	36		
	PrecMax	665	63	44	113	73	131	146	142	126	102	81	97	65		
	PrecMin	429	5	4	11	8	22	44	31	33	20	2	5	10		
	PrecStd	72	17	10	23	18	34	30	26	23	23	20	22	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	3.6	13.7	22.9	28.4	30.9	31.1	31.8	28.6	21.6	12.8	3.6		
		MCWB	1.1	1.0	8.7	12.4	16.0	21.4	21.8	21.6	18.9	14.0	8.0	0.9		
	2%	DB	0.2	1.5	8.2	18.9	24.8	27.5	29.1	28.8	25.4	17.8	8.7	0.8		
		MCWB	-0.9	0.2	4.6	10.1	14.1	18.6	21.0	20.2	17.3	11.5	5.2	-0.8		
	5%	DB	-2.8	-0.2	5.2	16.0	22.0	25.7	27.5	27.0	22.9	14.9	6.0	-1.1		
		MCWB	-3.9	-1.4	2.5	8.8	13.0	17.7	20.0	19.3	15.9	10.1	3.3	-2.1		
	10%	DB	-5.6	-3.3	2.9	12.8	19.2	23.8	25.9	25.2	20.4	12.3	3.5	-2.9		
		MCWB	-6.4	-4.5	1.0	7.0	11.5	16.6	19.0	18.3	14.9	9.0	1.7	-3.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.5	8.3	13.2	17.9	23.2	24.2	23.0	20.6	14.9	8.8	1.1		
		MCDB	2.7	2.9	11.8	20.8	24.9	28.2	29.0	29.0	25.5	20.2	11.8	3.1		
	2%	WB	-0.8	0.3	4.6	10.8	15.7	20.4	22.3	21.5	18.6	12.7	5.6	-0.5		
		MCDB	0.3	1.4	7.7	18.2	21.6	25.3	27.0	26.9	23.0	16.1	8.1	0.7		
	5%	WB	-3.7	-1.6	2.7	9.1	13.9	18.9	21.0	20.3	17.1	10.7	3.7	-2.0		
		MCDB	-2.7	-0.2	5.0	15.0	19.8	23.7	25.8	25.5	21.2	14.4	5.9	-1.1		
	10%	WB	-6.4	-4.5	1.1	7.4	12.5	17.6	19.8	19.2	15.6	8.8	1.7	-3.8		
		MCDB	-5.6	-3.4	2.7	12.7	18.4	22.1	24.5	23.8	19.7	12.1	3.5	-3.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.7	10.7	11.8	12.6	11.6	12.0	12.4	11.6	9.1	7.9	8.7		
		MCDBR	9.9	9.3	10.6	16.6	17.0	14.4	13.9	15.0	15.5	13.9	11.2	9.5		
	5% WB	MCWBR	9.0	8.2	7.7	9.5	8.8	7.5	7.3	7.5	8.5	8.5	8.0	8.5		
		MCWBR	9.9	9.1	9.8	15.4	13.8	12.2	12.4	13.0	12.8	12.3	10.3	9.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.259	0.283	0.307	0.354	0.364	0.380	0.391	0.373	0.342	0.297	0.264	0.242		
		taud	2.244	2.193	2.260	2.324	2.371	2.356	2.348	2.405	2.479	2.559	2.448	2.312		
	Edn at Noon	Edn at Noon	767	842	892	886	890	876	858	857	851	838	776	740		
		Edh at Noon	67	94	107	114	115	118	117	105	87	66	56	54		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.12	2.25	3.57	5.00	5.48	5.78	5.98	4.97	3.50	2.01	1.06	0.83		
	RadStd	RadStd	0.10	0.15	0.25	0.33	0.45	0.45	0.31	0.35	0.34	0.25	0.14	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	-0.58	-0.80	N/A	N/A	N/A	N/A

Nomenclature: See separate page